

# Saniat Javid Sohrawardi

Researcher in the fields of qualitative user studies, 6+ years of HCI and ML with 4+ years of industry experience in UXR, web development and design, and branding. Founder and lead of the DeFake Project focused research on the development and usability of deepfake detection and digital media forensics collaborating with journalists, intelligence analysts and law enforcement.

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John Sohrawardi

## EXPERIENCE

### Rochester Institute of Technology – Graduate Researcher

August 2018- October 2025 [expected] (Full time)

- **Lead** researcher and founder of the [DeFake project](#), growing the project into an interdisciplinary team of ~18 researchers collaborating across 4 universities.
- **Project manager** for the various sub-projects of the DeFake, handling tasks, goals, and project timelines across the team.
- Designed and carried out **qualitative user studies** with journalists and students with publications in [CHI](#), USENIX, SOUPS, [CCS](#), and [C+I](#)
- Designed interactive high fidelity **prototypes** in Figma to bolster ideation during **qualitative user studies** and test information flow.
- Developed **deep learning** models for deepfake detection, state of the art accuracies at the time of publication at [IJCB](#), [ISTSP](#).
- Effectively communicated research to audiences with varying expertise via **academic research**, blogs, **workshop/panel talks**, and interviews.
- [Currently](#), improving **usability** and **explainability** of deepfake detection and digital media forensics workflows for expert users.
- [Current](#) stage of dissertation involves developing an AI-assisted technique for training journalists in media verification.

### Boomerang Digital, Dhaka— Senior Web Developer

November 2014- August 2016 & August 2017- August 2018 (Full time)

- Collaborated on a **qualitative user research** study with Norman & Nielsen, analyzed the findings, and used them to design the national health portal user experience and interface.
- Designed **UX and responsive UI** for Telenor Health public health and lifestyle portal serving a user-base of ~173m.
- Designed **mobile UX/UI** for an Area 120 project, a job portal for South East Asia that created ~18k jobs in the first month.
- Designed UX and responsive UI for Telenor Health doctor call-centers (health informatics) increasing patient call efficiency.
- Designed mobile UX/UI for local business-to-distributor management platform launched nationwide in Bangladesh.
- Led and **mentored** a team of 4 developers on successful custom CMS-based web projects for local and foreign clients.
- Designed and developed websites with successful targets of 90+ performance scores on Google PageSpeed insights.

## SKILLS

### Research

Qualitative user studies, Usability testing, A/B testing & data analysis, Survey design & analysis, Prototyping

### Technical

**Programming:** JavaScript, Python, TypeScript, PHP, HTML, CSS, SQL  
**Machine Learning:** Lightning, Pytorch, Keras, TensorFlow  
**Web Development:** ReactJS, NextJS, ExpressJS, Flask, FeathersJS, Docker

### Design Skills

UX design (wireframing, information flow, interaction design), Responsive UI design, Prototyping (Figma, Moqups, Balsamiq), Visual communication (Adobe Creative Suite)

### Other Skills

Project management, Team leadership & mentorship, Academic writing & research publication, Workshop/panel presentations

## LANGUAGES

Fluent: English, Russian, Bengali

Intermediate: Ukrainian, Hindi

## Chaldal Limited, Dhaka — Software Engineer

August 2016 - June 2017 (Full time)

- Designed and developed online grocery website's coupon management page
- Designed and executed usability research on Chaldal's order making and scheduling platform to improve customer experience
- Deployed an **A/B testing** and analytics framework used to evaluate the website and the user experience of order scheduling pathways

## EDUCATION

### Rochester Institute of Technology, Rochester — PhD. in Computing & Information Sciences (Ongoing - 3.75/4)

Fall 2018 - Fall 2024/Spring 2025

*ESL Global Cybersecurity Institute* - Research in Human-Computer Interaction, Deep Learning, Media Forensics, and AI Ethics.

**Projects:** Deepfake Detection, Dungeons & Deepfakes: scenario-based evaluation of and training for journalistic deepfake detection workflows, Image-based Spear-phishing Attacks, Human behavior anomaly detection from videos using deep learning, GAN animated character generation

### North South University, Dhaka— BSc. in Computer Science and Engineering (3.65/4)

Fall 2009 - Spring 2014

**Minor in:** Genetics (Molecular Biology I & II, Microbial Genetics); **Electives:** Data Mining, Neural Networks, Quality Assurance, Parallel Processing; **Projects:** Browser-based Registration-free P2P Video Conferencing Platform, Bibliomining, ML-based user-submitted bug-classification

## PROJECTS

### DeFake Project — Deepfake Detection Platform - ongoing

Leading a deepfake detection project aimed at providing a tool for journalists to identify deepfake videos while providing them with interpretable information.

#### Sub-projects:

- *DeFake Analytic Interface*: **Qualitative studies** with journalists, fact-checkers, and intelligence analysts across the globe to understand requirements, current workflow, knowledge and evaluate prototypes. The goal of the project is to develop an intuitive, holistic media forensics experience that allows users to make well informed decisions based on analytic response.
- *DeFake App*: Development, implementation and deployment of ML toolsets and front and back-end of the application.
- *DeFake DnD*: Innovative deepfake detection usability study to understand risks, accessibility, and explanation needs for a deepfake detection tool to make a positive impact on workflow. Study uses a scenario-based role-play procedure.

## CERTIFICATIONS

NVIDIA: Conversational AI, Transformers, Generative AI

Knight: Advanced digital investigations

CITI: Social & Behavioral Research

## PUBLICATIONS

Citations: 389 (May 2025)

**Understanding and Empowering Intelligence Analysts: User-Centered Design for Deepfake Detection Tools**  
2025 - ACM CHI

**Dungeons & Deepfakes: Using scenario-based role-play to study journalists' behavior towards using AI-based verification tools for video content**  
2024 - ACM CHI

**A First Look into Targeted Clickbait and its Countermeasures: The Power of Storytelling**  
2024 - ACM CHI

**Recurrent Convolutional Structures for Audio Spoof and Video Deepfake Detection**  
2020 - IEEE JSTSP

**Leveraging edges and optical flow on faces for deepfake detection**  
2020 - IEEE IJCB

**DeFaking Deepfakes: Understanding Journalists Needs**  
2020 - C+J, SOUPS

- *We.DeFake*: A **federated social media** platform hosted over ActivityPub for a collaborative community of experts, researchers, fact-checkers, intelligence analysts, and journalists against manipulated multimedia.

**Tools:** Figma, Python Flask, ReactJS, Pytorch, Keras, Pleroma, Soapbox

**Concepts:** Deep Learning, Usable Security, Media Forensics, User Experience, Collaborative Computing, Social Media, Web Development

**Portal:** <https://defake.app/about>, <https://we.defake.app>

### **DartSessions — P2P Video Conferencing Platform - 2013**

Built a P2P Registration-free Video Conferencing Platform that showcased a highly intuitive and user friendly interface.

**Tools:** WebRTC, NodeJS, PHP, HTML, CSS and JavaScript

### **Tonic — User Experience and User Interface Design - 2015**

Carried out physical User Experience testing, developed wireframes and Desktop and Mobile UI designs for Telnor Health's Tonic Health Portal used across the nation ([read article](#)).

**Tools:** Figma, Moqups, Canvasflip, Balsamiq

### **CarrotCake — User Experience and User Interface Design - 2015**

Carried out User Experience research and testing, developed wireframes and User Interface designs for Telnor Health's Carrotcake Doctors' Patient Data Input Interface project at Boomerang Digital.

**Tools:** Figma, Balsamiq, Illustrator

### **Towards Robust Open-World Detection of Deepfakes**

2019 - CCS

### **A comparative study of text classification algorithms on user submitted bug reports 2014 -**

IEEE

### **Bibliomining on North South University library data**

2013 - IEEE

## **MEMBERSHIPS**

Scientific Working Group on Digital Evidence (SWGDE)

AI Risk and Vulnerability Alliance (AVID)

Association for Computing Machinery (ACM)

IEEE Biometrics Council

## **TEACHING**

Cyber Analytics & ML (Spring'25)